

MINISTRY OF HOME AFFAIRS | BUREAU OF IMMIGRATION
GOVERNMENT OF INDIA | BORDER SECURITY DIRECTORATE

VeriDoc AI

AI-Based Fake Identity & Document Screening System
for Government & Border Security

SMART INDIA HACKATHON 2026

Problem Statement SIH26188

AI-Based Fake Identity & Document Screening for Government & Border Security

Team VeriDoc

Project Dossier | Version 1.0 | September 2026
Demonstration build - all analytics data is simulated

1. Executive Overview

VeriDoc AI is an AI-based fake identity and document screening system built for the Ministry of Home Affairs and border-security command centres. It ingests a scan or photograph of an identity document - Aadhaar Card, PAN Card, Passport, or Voter ID - and runs a forensic pipeline of Optical Character Recognition (OCR), typographic font-consistency analysis, Error-Level Analysis (ELA), and face-biometric comparison. The fused outputs collapse into a single, explainable authenticity score with an immediate PASS / REVIEW / REJECT decision.

The problem

Forged and altered identity documents are a primary vector for illegal immigration, identity fraud, and benefits misuse. Manual inspection at checkpoints is slow, inconsistent, and easily defeated by high-quality forgeries. Verification officers need a fast, objective, and auditable screening tool that works at the point of verification.

The solution

- Instant forensic verdict - a 0-100% authenticity score with colour-coded risk, computed in under 1.2 seconds.
- Explainable results - every score is backed by inspectable tamper zones rendered directly on the document image.
- Pre-flight structural screening - rejects non-ID uploads (wallpapers, photos, memes) before analysis is attempted.
- End-to-end audit trail - every scan, including rejections, is appended to a searchable, append-only log.
- Operator and analyst dashboards - live registry, tamper analytics, and CSV export for command-centre oversight.

Key metrics (seeded operational baseline)

Documents screened	12,847
Counterfeits flagged	1,203 (9.4% of scans)
Average decision time	1.12 s (target < 1.2 s)
Model AUC	99.2%
Document types verified	Aadhaar - PAN - Passport - Voter ID
Deployment footprint	128 MCP checkpoints, 24x7 uptime

Technology stack

React 19 - Vite - Tailwind CSS 4 - Node.js 24 - VeriDoc forensic engine v2.4.1. The demonstration build ships a high-fidelity simulated engine so the full operator workflow can be exercised without external ML services.

2. System Architecture

The system is a layered, browser-first architecture. Every component below is implemented and running in the build; the engine layer is a high-fidelity simulator ready for the production ML service.



Component responsibilities

- Presentation layer (React 19 + Tailwind 4): a five-view single-page app - Scan, Inspection, Dashboard, Analytics, Audit - styled as a dark command-centre interface.
- Upload & pre-flight: drag-and-drop accepts PNG, JPG and PDF; the check runs the moment a file lands, decoding image dimensions and scanning the filename for non-ID markers.
- Engine pipeline: four sequential phases (OCR, font consistency, ELA, scoring), each surfaced to the operator with live progress.
- Results & evidence: bounding boxes localise anomalies ("Mismatched Font on DOB", "Altered Profile Photo Edges"); a metadata card shows fields and face-match status; a radial gauge renders the score.
- Data layer: a seeded operational dataset plus session-scoped records - every completed scan lands in the register and audit log in real time.
- Reporting: per-document PDF reports and CSV export of the audit trail are generated client-side, with no server round-trip.

Security posture

- AES-256 encryption at rest and TLS 1.3 in transit.
- Append-only audit ledger - each entry records operator, terminal, IP and action.
- Zero-retention pipeline - document images are not persisted after analysis in the demo.
- Role labels on every action (e.g. "Operator - Live Console") for accountability.

3. Screen Descriptions

The build is a fully working browser demo: upload a document (or load a sample), run the forensic analysis, and watch the result appear in the inspection view, dashboard register and audit log in real time.

3.1 Scanner / Landing screen

- Brand bar "VeriDoc AI | MHA Portal" with governance badges: Ministry of Home Affairs, Bureau of Immigration, SIH 2026.
- Drag-and-drop upload zone accepting PNG, JPG and PDF files.
- Document type selector: Aadhaar Card, PAN Card, Passport, Voter ID.
- Simulation mode selector: Auto-Detect, Force Genuine (Pass), Force Counterfeit / Tampered (Fail), Force Invalid Document (Reject).
- One-click sample loads (genuine and tampered) for demonstrations.
- "Run Forensic AI Analysis" launches the four-phase pipeline with live progress: Optical Character Recognition... -> Checking Font Consistency... -> Running Error Level Analysis (ELA)... -> Calculating Authenticity Trust Score.
- A non-ID upload is flagged instantly with a red chip and its reason (e.g. "Filename signal 'cat'", "Atypical aspect ratio 4.29:1").

3.2 Inspection Results & Verification view

- Side-by-side layout: the document preview sits on the left, the extracted metadata card on the right.
- Interactive bounding boxes overlay the document, each labelled with its finding (e.g. "Mismatched Font on DOB", "Altered Profile Photo Edges", "MRZ Ink Bleed / Re-print") and a severity colour.
- Metadata card fields: Full Name, Document Number, Issue Date, Validity, Face-Match status (PASSED / FAILED), Tamper Likelihood, processing time.
- Radial trust gauge scored 0-100%, colour-coded by risk band.
- A critical failure renders at 28% with a pulsing banner: "CRITICAL TAMPER DETECTED: Spliced photo & font mismatch".
- Rejected uploads render a dedicated screen at 0% with status "INVALID DOCUMENT / UNRECOGNIZED FORMAT".
- "Download Forensic PDF Report" - a per-inspection evidence pack (also available as a rejection report).

Risk bands:

 75-100% Emerald - Authentic

 50-74% Amber - Suspicious / review

 0-49% Red - Counterfeit / reject

3. Screen Descriptions (continued)

3.3 Verification Dashboard

- Metric cards: Total Scanned (12,847), Counterfeits Flagged (1,203), Average Processing Time (1.12 s), plus a live "+N THIS SESSION" chip.
- Searchable register of past verifications: Document ID, Type, Timestamp, Holder, Risk level (Authentic / Suspicious / Counterfeit / Rejected), and a per-row "Download Forensic PDF Report" action.
- Risk filter chips (All / Authentic / Suspicious / Counterfeit / Rejected) for fast triage.
- Session-scoped rows are tagged SESSION, styled distinctly, and pinned to the top of the register.

3.4 Tamper Analytics

- Donut chart of verdict distribution (Authentic vs Suspicious vs Counterfeit).
- Bar chart of tamper frequency by document type (Aadhaar, PAN, Passport, Voter ID).
- Trend line of detection rate over time for command-centre monitoring.
- Top tamper techniques leaderboard - photo splicing, font mismatch, MRZ re-print, ghost overlay.

3.5 Audit Logs

- Append-only, searchable event trail: operator, terminal, IP, action, severity, timestamp.
- Distinct entries for completed scans and for pre-screen rejections ("Scan rejected at structural pre-screen").
- Severity badges and CSV export of the full trail, including live session entries.

Shared behaviour

- Dark, high-contrast cybersecurity aesthetic: deep navy/slate background, crisp white typography, alert accents - emerald for verified, amber for review, crimson for counterfeit.
- Fully responsive: the operator views collapse into a mobile navigation menu; tables and charts remain usable at checkpoint-tablet sizes.
- All numbers in this dossier (except live session rows) come from the seeded operational baseline dataset.

4. Development, Roadmap & Closing

Running the build

- npm install - installs Vite, React, Tailwind CSS 4 and tooling.
- npm run dev - starts the Vite development server (default http://localhost:5173).
- npm run build - produces the production bundle (verified: 40 modules, clean compile).
- node generate-dossier.mjs - regenerates this dossier PDF.

Roadmap to production

- Swap the simulated engine for real inference: OCR (e.g. Tesseract), font forensics, ELA and face biometrics behind a hardened API.
- Render uploaded PDF pages client-side with pdf.js so multi-page passports are analysable.
- Persist the register and audit ledger to a backend database with role-based access control.
- Camera-capture kiosk mode for checkpoint terminals, with NFC / QR authenticity binding.

Closing statement

VeriDoc AI demonstrates a complete, operator-ready workflow for AI-based identity-document screening: a fast forensic pipeline, explainable tamper evidence, live command-centre analytics, and a tamper-evident audit trail. Every screen described in this dossier is implemented and running in the demonstration build, and each scan performed during a session flows end-to-end from the scanner to the dashboard register and audit log.

Prepared for evaluation at Smart India Hackathon 2026 - Problem SIH26188. Contact the team for a live walkthrough, source access, or a deeper technical brief on the forensic pipeline.